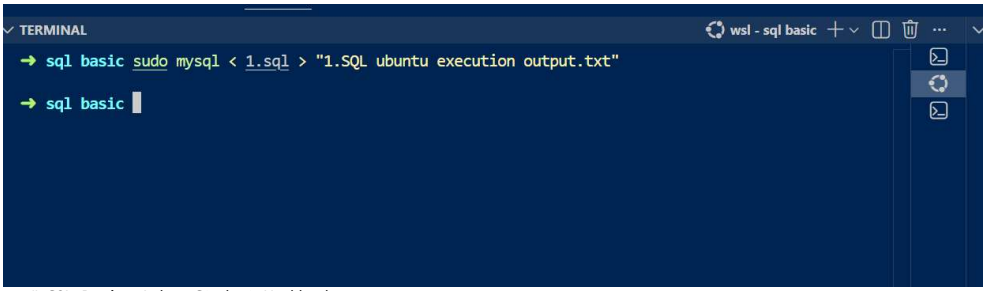
calibo




```
-- # SQL Basics Lab - Student Workbook
-- ## Lab Overview
-- In this lab, you will practice SQL SELECT and WHERE statements using a
company database. The database contains information about employees, products,
and orders. Work through each exercise sequentially and write your SQL queries
to solve the problems.
-- in ubuntu terminal to access mysql
-- open ubuntu app and run
-- sudo mysql
-- password
-- it open sql shell

-- ## Database Schema
-- You will work with three tables:
-- **employees**
-- - employee_id (INT, Primary Key)
-- - first_name (VARCHAR)
-- - last_name (VARCHAR)
-- - email (VARCHAR)
-- - department (VARCHAR)
-- - job_title (VARCHAR)
-- - salary (DECIMAL)
-- - hire_date (DATE)
-- - manager_id (INT)
-- - city (VARCHAR)
-- CREATE DATABASE basicsqltable;
USE basicsqltable;
SELECT 'open ubuntu app and run sudo mysql password it open sql shell';
-- select 'sudo mysql basicsqltable < 1.sql';
drop table employees;
drop table products;
drop table orders;
-- delete from employees;

CREATE TABLE employees(
employee_id INT,
first_name VARCHAR(50),
last_name VARCHAR(50),
email VARCHAR(100),
department VARCHAR(50),
job_title VARCHAR(50),
salary DECIMAL,
hire_date DATE,
manager_id INT,
city VARCHAR(50),
PRIMARY KEY(employee_id)
);


-- **products**
-- - product_id (INT, Primary Key)
-- - product_name (VARCHAR)
-- - category (VARCHAR)
-- - price (DECIMAL)
-- - stock_quantity (INT)
-- - supplier (VARCHAR)
CREATE TABLE products(
product_id INT PRIMARY KEY,
product_name VARCHAR(100),
category VARCHAR(50),
price DECIMAL(10,2),
stock_quantity INT,
supplier VARCHAR(50)
);
-- **orders**
-- - order_id (INT, Primary Key)
-- - customer_name (VARCHAR)
-- - product_id (INT)
-- - quantity (INT)
-- - order_date (DATE)
-- - total_amount (DECIMAL)
-- - status (VARCHAR)
CREATE TABLE orders(
order_id INT PRIMARY KEY,
customer_name VARCHAR(100),
product_id INT,
quantity INT,
order_date DATE,
total_amount DECIMAL(10,2),
status VARCHAR(50)
);
SHOW TABLES;

-- ## Instructions
```

```
-- 1. Read each exercise carefully
-- 2. Write your SQL query below each exercise
-- 3. Test your query to ensure it returns the expected results
-- 4. Verify the number of rows returned matches the expected result
-- 5. Ask for help if you're stuck
-- ---

-- ## Part 1: Basic SELECT Queries
-- ### Exercise 1.1: Select All Columns
-- **Task**: Retrieve all information from the employees table.
-- **Expected Result**: 15 rows with all columns

-- ```
select 'exercise 1.1';
DELETE FROM employees;
INSERT INTO employees VALUES
(1,'John','Sharma','john@company.com','IT','Dev',90000,'2020-01-10',NULL,'San
Francisco'),
(2,'James','Johnson','james@company.com','IT','Dev',95000,'2019-02-12',1,'San
Francisco'),
(3,'Jack','Singh','jack@company.com','IT','Lead',92000,'2019-03-15',1,'San
Francisco'),
(4,'Julia','Thomas','julia@company.com','IT','Tester',72000,'2019-04-18',1,'New
York'),
(5,'Nea','Gupta','neha@company.com','IT','Tester',72000,'2019-05-20',1,'Chicago
'),
(6,'Rohit','Wilson','rohit@company.com','Sales','Exec',60000,'2019-06-10',NULL,
'Boston'),
(7,'Pooja','Mehta','pooja@company.com','Sales','Mgr',85000,'2019-07-12',6,'Delh
i'),
(8,'Suresh','Rao','suresh@company.com','Sales','Exec',55000,'2020-08-12',6,'New
York'),
(9,'Kiran','Das','kiran@company.com','Marketing','Mgr',74000,'2020-09-15',NULL,
'Boston'),
(10,'Ritika','Nair','ritu@company.com','Marketing','Exec',60000,'2020-10-18',9,
'Chicago'),

(11,'Manoj','Iyer','manoj@company.com','HR','Mgr',58000,'2020-11-20',1,'Boston'
),
(12,'Sneha','Anderson','sneha@company.com','HR','Exec',52000,'2021-01-05',11,'D
elhi'),
(13,'Vikas','Jain','vikas@company.com','Finance','Acc',61000,'2021-02-10',1,'Ch
icago'),
(14,'Arun','Malik','tarun@company.com','Finance','Acc',60000,'2021-03-12',13,'B
oston'),
(15,'Anita','Roy','anita@company.com','Finance','Rep',48000,'2021-04-15',1,'New
York');

DELETE FROM products;
INSERT INTO products VALUES
(1,'USB Cable','Electronics',150,120,'TechCorp'),
(2,'Headphones','Electronics',850,40,'TechCorp'),
(3,'Keyboard','Electronics',650,30,'CableWorks'),
(4,'Mouse','Electronics',80,35,'CableWorks'),
(5,'Laptop Bag','Accessories',1200,60,'BagWorld'),
(6,'Mobile Cover','Accessories',250,45,'TechCorp'),
(7,'Charger','Electronics',90,30,'TechCorp'),
(8,'Power Bank','Electronics',1500,40,'PowerHub'),
(9,'Pen Drive','Electronics',600,20,'TechCorp'),
(10,'Notebook','Stationery',60,200,'PaperMart'),
(11,'Pen','Stationery',20,0,'PaperMart'),
(12,'Water Bottle','Accessories',300,70,'BottleCo'),
(13,'Earphones','Electronics',450,80,'CableWorks'),
(14,'Calculator','Electronics',900,10,'TechCorp'),
(15,'School Bag','Accessories',1800,22,'BagWorld');

DELETE FROM orders;
INSERT INTO orders VALUES
(1,'Amit',1,3,'2024-03-05',300,'Processing'),
(2,'Bina',2,2,'2024-03-10',250,'Shipped'),
(3,'Chetan',3,2,'2024-03-15',220,'Delivered'),
(4,'Divya',4,1,'2024-03-20',210,'Processing'),
(5,'Esha',5,2,'2024-02-12',260,'Delivered'),
(6,'Farhan',6,1,'2024-02-25',230,'Pending'),
(7,'Gopal',7,2,'2024-04-02',240,'Shipped'),
(8,'Harish',8,1,'2024-01-18',260,'Pending'),
(9,'Isha',9,1,'2024-01-22',150,'Pending'),
(10,'Jay',10,1,'2024-04-05',100,'Cancelled');
SELECT * FROM employees;
```

```
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➜ sql basic sudo mysql basicsqltable < 1.sql

open ubuntu app and run sudo mysql password it open sql shell
open ubuntu app and run sudo mysql password it open sql shell
Tables_in_basicsqltable
employees
orders
products
exercise 1.1
exercise 1.1
employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Sharma	john@company.com	IT	Dev	90000	2020-01-10	NULL	San Francisco
2	James	Johnson	james@company.com	IT	Dev	95000	2019-02-12	1	San Francisco
3	Jack	Singh	jack@company.com	IT	Lead	92000	2019-03-15	1	San Francisco
4	Julia	Thomas	julia@company.com	IT	Tester	72000	2019-04-18	1	New York
5	Nea	Gupta	neha@company.com	IT	Tester	72000	2019-05-20	1	Chicago
6	Rohit	Wilson	rohit@company.com	Sales	Exec	60000	2019-06-10	NULL	Boston
7	Pooja	Mehta	pooja@company.com	Sales	Mgr	85000	2019-07-12	6	Delhi
8	Suresh	Rao	suresh@company.com	Sales	Exec	55000	2020-08-12	6	New York
9	Kiran	Das	kiran@company.com	Marketing	Mgr	74000	2020-09-15	NULLBoston
10	Ritika	Nair	ritu@company.com	Marketing	Exec	60000	2020-10-18	9	Chicago
11	Manoj	Iyer	manoj@company.com	HR	Mgr	58000	2020-11-20	1	Boston
12	Sneha	Anderson	sneha@company.com	HR	Exec	52000	2021-01-05	11 Delhi
13	Vikas	Jain	vikas@company.com	Finance Acc	61000	2021-02-10	1	Chicago
14	Arun	Malik	tarun@company.com	Finance Acc	60000	2021-03-12	13	Boston
15	Anita	Roy	anita@company.com	Finance Rep	48000	2021-04-15	1	New York
```

```
-- ### Exercise 1.2: Select Specific Columns
-- **Task**: Retrieve only the first name, last name, and email of all
employees.
-- **Expected Result**: 15 rows with 3 columns

select 'exercise 1.2';
select first_name, last_name, email from employees;
```

```
exercise 1.2
first_name	last_name	email
John	Sharma	john@company.com
James	Johnson	james@company.com
Jack	Singh	jack@company.com
Julia	Thomas	julia@company.com
Nea	Gupta	neha@company.com
Rohit	Wilson	rohit@company.com
Pooja	Mehta	pooja@company.com
Suresh	Rao	suresh@company.com
Kiran	Das	kiran@company.com
Ritika	Nair	ritu@company.com
Manoj	Iyer	manoj@company.com
Sneha	Anderson	sneha@company.com
Vikas	Jain	vikas@company.com
Arun	Malik	tarun@company.com
Anita	Roy	anita@company.com
```

```
-- ---

-- ### Exercise 1.3: Using Column Aliases
-- **Task**: Retrieve employee first name as "First Name", last name as "Last
Name", and salary as "Annual Salary".
-- **Expected Result**: 15 rows with renamed column headers

-- ```
select 'exercise 1.3';
select first_name as 'First Name', last_name as 'Last Name', salary as 'Annual
Salary' from employees;
-- ---
```

```
exercise 1.3
First Name	Last Name	Annual Salary
John	Sharma	90000
James	Johnson	95000
Jack	Singh	92000
Julia	Thomas	72000
Nea	Gupta	72000
Rohit	Wilson	60000
Pooja	Mehta	85000
Suresh	Rao	55000
Kiran	Das	74000
Ritika	Nair	60000
Manoj	Iyer	58000
Sneha	Anderson	52000
Vikas	Jain	61000
Arun	Malik	60000
Anita	Roy	48000
```

```
-- ### Exercise 1.4: SELECT DISTINCT
-- **Task**: Find all unique departments in the company.
-- **Expected Result**: 5 unique departments

select 'exercise 1.4';
SELECT DISTINCT department FROM employees;
```

```
exercise 1.4
department
IT
Sales
Marketing
HR
Finance
```

```
-- ---
-- ### Exercise 1.5: SELECT DISTINCT with Multiple Columns
-- **Task**: Find all unique combinations of department and city.
-- **Expected Result**: Multiple unique combinations


select 'exercise 1.5';
SELECT DISTINCT department,city FROM employees;
```

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```
exercise 1.5
department      city
IT      San Francisco
IT      New York
IT      Chicago
Sales   Boston
Sales   Delhi
Sales   New York
Marketing       Boston
Marketing       Chicago
HR      Boston
HR      Delhi
Finance Chicago
Finance Boston
Finance New York
```

```
-- ---

-- ## Part 2: Expressions in SELECT
-- ### Exercise 2.1: Arithmetic Operations
-- **Task**: Calculate the monthly salary for each employee (annual salary
divided by 12).
-- **Expected Result**: 15 rows showing both annual and monthly salary

SELECT 'Exercise 2.1: monthly salary'AS "";
SELECT first_name,last_name,salary,salary/12 AS monthly_salary FROM employees;
-- ---
```

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```
Exercise 2.1: monthly salary
first_name      last_name       salary  monthly_salary
John    Sharma  90000   7500.0000
James   Johnson 95000   7916.6667
Jack    Singh   92000   7666.6667
Julia   Thomas  72000   6000.0000
Nea     Gupta   72000   6000.0000
Rohit   Wilson  60000   5000.0000
Pooja   Mehta   85000   7083.3333
Suresh  Rao     55000   4583.3333
Kiran   Das     74000   6166.6667
Ritika  Nair    60000   5000.0000
Manoj   Iyer    58000   4833.3333
Sneha   Anderson        52000   4333.3333
Vikas   Jain    61000   5083.3333
Arun    Malik   60000   5000.0000
Anita   Roy     48000   4000.0000
```

```
-- ### Exercise 2.2: Multiple Calculations
-- **Task**: Display employee names with their annual salary, monthly salary,
and weekly salary (annual/52).
-- **Expected Result**: 15 rows with salary broken down by different time
periods


SELECT 'Exercise 2.2:' AS "";
SELECT first_name,last_name,salary,salary/12 AS monthly_salary,salary/52 AS
weekly_salary FROM employees;
-- ```
```

```
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Exercise 2.2:
first_name      last_name       salary  monthly_salary  weekly_salary
John    Sharma  90000   7500.0000       1730.7692
James   Johnson 95000   7916.6667       1826.9231
Jack    Singh   92000   7666.6667       1769.2308
Julia   Thomas  72000   6000.0000       1384.6154
Nea     Gupta   72000   6000.0000       1384.6154
Rohit   Wilson  60000   5000.0000       1153.8462
Pooja   Mehta   85000   7083.3333       1634.6154
Suresh  Rao     55000   4583.3333       1057.6923
Kiran   Das     74000   6166.6667       1423.0769
Ritika  Nair    60000   5000.0000       1153.8462
Manoj   Iyer    58000   4833.3333       1115.3846
Sneha   Anderson        52000   4333.3333       1000.0000
Vikas   Jain    61000   5083.3333       1173.0769
Arun    Malik   60000   5000.0000       1153.8462
Anita   Roy     48000   4000.0000       923.0769
```

```
-- ---
-- ### Exercise 2.3: String Concatenation
-- **Task**: Create a full name by combining first name and last name with a
space in between.
-- **Hint**: Use the appropriate concatenation method for your database (|| for
PostgreSQL/Oracle, CONCAT for MySQL)
-- **Expected Result**: 15 rows with combined full names


select 'Exercise 2.3:' AS "";
select concat(first_name,' ',last_name) as full_name from employees;
```

```
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Exercise 2.3:
full_name
John Sharma
James Johnson
Jack Singh
Julia Thomas
Nea Gupta
Rohit Wilson
Pooja Mehta
Suresh Rao
Kiran Das
Ritika Nair
Manoj Iyer
Sneha Anderson
Vikas Jain
Arun Malik
Anita Roy
```

```
-- ## Part 3: WHERE Clause - Basic Filtering
-- ### Exercise 3.1: Equality Condition
-- **Task**: Find all employees who work in the IT department.
-- **Expected Result**: 5 employees


select 'Exercise 3.1:'AS "";
select * from employees where department='IT';
```

```
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Exercise 3.1:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
```

```
-- ### Exercise 3.2: Numeric Comparison
-- **Task**: Find all employees with a salary greater than 70000.
-- **Expected Result**: 8 employees(9 employees (based on current dataset)


select 'Exercise 3.2:' AS "";
select * from employees where salary>70000;
```

```
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TERMINAL                                                    wsl - sql basic

Exercise 3.2:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
```

```
-- ---
-- ### Exercise 3.3: Not Equal To
-- **Task**: Find all employees who are NOT in the Sales department.
```

```
-- **Expected Result**: 10 employees
```

```
select 'Exercise 3.3:' AS "";
select * from employees where department<>'Sales';
```

```
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TERMINAL

Exercise 3.3:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id
city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
12      Sneha   Anderson        sneha@company.com       HR      Exec    52000   2021-01-05      11 Delhi
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ### Exercise 3.4: Less Than or Equal To
-- **Task**: Find all employees with a salary of 60000 or less.
-- **Expected Result**: 7 employees
```

```
select 'Exercise 3.4:' AS "";
select * from employees where salary<=60000;
```

```
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TERMINAL

Exercise 3.4:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id
city
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
12      Sneha   Anderson        sneha@company.com       HR      Exec    52000   2021-01-05      11 Delhi
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ## Part 4: Logical Operators
-- ### Exercise 4.1: AND Operator
-- **Task**: Find all employees in the IT department who earn more than 85000.
-- **Expected Result**: 3 employees
```

```
select 'Exercise 4.1:'AS "";
select * from employees where department='IT' and salary>85000;
```

```
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TERMINAL                                                    wsl - sql basic

Exercise 4.1:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id
city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
```

```
-- ### Exercise 4.2: OR Operator
-- **Task**: Find all employees who work in either Sales or Marketing
departments.
-- **Expected Result**: 8 employees
```

```
select 'Exercise 4.2:' AS "";
select * from employees where department='Sales' or department='Marketing';
```

```
DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS
TERMINAL                                                    wsl - sql basic

Exercise 4.2:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_
id      city
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
```

```
-- ### Exercise 4.3: Combining AND and OR
-- **Task**: Find employees in Sales or Marketing departments who earn more
than 60000.
-- **Hint**: Use parentheses to group the OR conditions
-- **Expected Result**: 2 employees
```

```
select 'Exercise 4.3:' AS "";
select * from employees where (department='Sales' or department='Marketing')
and salary>60000;
```

```
DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS
TERMINAL                                                    wsl - sql basic

Exercise 4.3:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_
id      city
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
```

```
-- ### Exercise 4.4: NOT Operator
-- **Task**: Find all employees who are NOT in IT or HR departments.
-- **Expected Result**: 8 employees
```

```
select 'Exercise 4.4:' AS "";
select * from employees where  department<>'IT' and department<>'HR';
select 'Exercise 4.4:' AS "";
select "can also be wriiten as" AS "";
select * from employees where not(department='IT' or department='HR');
```

```
Exercise 4.4:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_i
d       city
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York

Exercise 4.4:

can also be wriiten as
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_i
d       city
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ## Part 5: Special WHERE Conditions
-- ### Exercise 5.1: BETWEEN Operator
-- **Task**: Find all employees with salaries between 55000 and 75000
(inclusive).
-- **Expected Result**: 9 employees
```

```
select 'Exercise 5.1:' AS "";
select * from employees where salary between 55000 and 75000;
select 'Exercise 5.1:' AS "";
select 'can also be written as' AS "";
select * from employees where salary>=55000 and salary<=75000;
```

```
Exercise 5.1:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston

Exercise 5.1:

can also be written as
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
```

```
-- ### Exercise 5.2: IN Operator
-- **Task**: Find all employees located in New York, Chicago, or Boston.
-- **Expected Result**: 9 employees
```

```
select 'Exercise 5.2:' AS "";
select * from employees where city in('New York','Chicago','Boston');
```

```
DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS
TERMINAL                                                    wsl - sql basic

Exercise 5.2:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ### Exercise 5.3: LIKE with % Wildcard
-- **Task**: Find all employees whose first name starts with 'J'.
-- **Expected Result**: 4 employees
```

```
select 'Exercise 5.3:' AS "";
select * from employees where first_name like 'J%';
```

```
DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS
TERMINAL                                                    wsl - sql basic

Exercise 5.3:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
```

```
-- ### Exercise 5.4: LIKE with Multiple Patterns
-- **Task**: Find all employees whose last name ends with 'son'.
-- **Expected Result**: 3 employees
```

```
select 'Exercise 5.4:' AS "";
select * from employees where last_name like '%son';
```

```
TERMINAL                                                    wsl - sql basic

Exercise 5.4:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
12      Sneha   Anderson        sneha@company.com       HR      Exec    52000   2021-01-05      11 Delhi
```

```
-- ### Exercise 5.5: LIKE with _ Wildcard
-- **Task**: Find all employees whose first name is exactly 4 letters long.
-- **Hint**: Use four underscore characters
-- **Expected Result**: 3 employees
```

```
select 'Exercise 5.5:' AS "";
select * from employees where first_name like '____';
```

```
DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS
TERMINAL                                                    wsl - sql basic

Exercise 5.5:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston

Exercise 5.6:
```

```
-- ### Exercise 5.6: IS NULL
-- **Task**: Find all employees who have no manager (manager_id is NULL).
-- **Hint**: Do not use = with NULL
-- **Expected Result**: 3 employees
```

```
select 'Exercise 5.6:' AS "";
select * from employees where manager_id is null;
```

```
TERMINAL                                                                wsl - sql basic

Exercise 5.6:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
```

```
-- ### Exercise 5.7: IS NOT NULL
-- **Task**: Find all employees who have a manager.
-- **Expected Result**: 12 employees


select 'Exercise 5.7:' AS "";
select * from employees where manager_id is not null;
```

```
TERMINAL                                                                wsl - sql basic

Exercise 5.7:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
12      Sneha   Anderson        sneha@company.com       HR      Exec    52000   2021-01-05      11 Delhi
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ## Part 6: Working with Products Table
-- ### Exercise 6.1: Low Stock Alert
-- **Task**: Find all products with stock quantity less than 50.
-- **Expected Result**: 8 products


select 'Exercise 6.1:' as "";
select * from products where stock_quantity<50;
```

```
TERMINAL

Exercise 6.1:
product_id      product_name    category        price   stock_quantity  supplier
2       Headphones      Electronics     850.00  40      TechCorp
3       Keyboard        Electronics     650.00  30      CableWorks
4       Mouse   Electronics     80.00   35      CableWorks
6       Mobile Cover    Accessories     250.00  45      TechCorp
7       Charger Electronics     90.00   30      TechCorp
8       Power Bank      Electronics     1500.00 40      PowerHub
9       Pen Drive       Electronics     600.00  20      TechCorp
11      Pen     Stationery      20.00   0       PaperMart
14      Calculator      Electronics     900.00  10      TechCorp
15      School Bag      Accessories     1800.00 22      BagWorld
```

```
-- ### Exercise 6.2: Out of Stock
-- **Task**: Find all products that are completely out of stock (stock quantity
equals 0).
-- **Expected Result**: 1 product


select 'Exercise 6.2:' as "";
select * from products where stock_quantity=0;
```

```
TERMINAL

Exercise 6.2:
product_id      product_name    category        price   stock_quantity  supplier
11      Pen     Stationery      20.00   0       PaperMart
```

```
-- ### Exercise 6.3: Price Range
-- **Task**: Find all products priced between 20 and 100 dollars.
-- **Expected Result**: 7 products


select 'Exercise 6.3:' as "";
select * from products where price between 20 and 100;
select 'can also be written as' as "";
select * from products where price>=20 and price<=100;
```

```
TERMINAL

Exercise 6.3:
product_id      product_name    category        price   stock_quantity  supplier
4       Mouse   Electronics     80.00   35      CableWorks
7       Charger Electronics     90.00   30      TechCorp
10      Notebook        Stationery      60.00   200     PaperMart
11      Pen     Stationery      20.00   0       PaperMart

can also be written as
product_id      product_name    category        price   stock_quantity  supplier
4       Mouse   Electronics     80.00   35      CableWorks
7       Charger Electronics     90.00   30      TechCorp
10      Notebook        Stationery      60.00   200     PaperMart
11      Pen     Stationery      20.00   0       PaperMart
```

```
-- ### Exercise 6.4: Specific Categories
-- **Task**: Find all products in Electronics or Accessories categories.
-- **Expected Result**: 13 products
```

```
select 'Exercise 6.4:' as "";
select * from products where category in('Electronics','Accessories');
```

```
TERMINAL

Exercise 6.4:
product_id      product_name    category        price   stock_quantity  supplier
1       USB Cable       Electronics     150.00  120     TechCorp
2       Headphones      Electronics     850.00  40      TechCorp
3       Keyboard        Electronics     650.00  30      CableWorks
4       Mouse   Electronics     80.00   35      CableWorks
5       Laptop Bag      Accessories     1200.00 60      BagWorld
6       Mobile Cover    Accessories     250.00  45      TechCorp
7       Charger Electronics     90.00   30      TechCorp
8       Power Bank      Electronics     1500.00 40      PowerHub
9       Pen Drive       Electronics     600.00  20      TechCorp
12      Water Bottle    Accessories     300.00  70      BottleCo
13      Earphones       Electronics     450.00  80      CableWorks
14      Calculator      Electronics     900.00  10      TechCorp
15      School Bag      Accessories     1800.00 22      BagWorld
```

```
-- ### Exercise 6.5: Supplier Search
-- **Task**: Find all products supplied by TechCorp.
-- **Expected Result**: 6 products
```

```
select 'Exercise 6.5:' as "";
select * from products where supplier='TechCorp';
```

```
TERMINAL

Exercise 6.5:
product_id      product_name    category        price   stock_quantity  supplier
1       USB Cable       Electronics     150.00  120     TechCorp
2       Headphones      Electronics     850.00  40      TechCorp
6       Mobile Cover    Accessories     250.00  45      TechCorp
7       Charger Electronics     90.00   30      TechCorp
9       Pen Drive       Electronics     600.00  20      TechCorp
14      Calculator      Electronics     900.00  10      TechCorp
```

```
-- ## Part 7: Working with Orders Table
-- ### Exercise 7.1: Recent Orders
-- **Task**: Find all orders placed in March 2024.
-- **Hint**: Use BETWEEN with dates in 'YYYY-MM-DD' forma
-- **Expected Result**: 4 orders
```

```
select 'Exercise 7.1:' as "";
select * from orders where order_date between '2024-03-01' and '2024-03-31';
```

```
TERMINAL

Exercise 7.1:
order_id        customer_name   product_id      quantity        order_date      total_amount    status
1       Amit    1       3       2024-03-05      300.00  Processing
2       Bina    2       2       2024-03-10      250.00  Shipped
3       Chetan  3       2       2024-03-15      220.00  Delivered
4       Divya   4       1       2024-03-20      210.00  Processing
```

```
-- ### Exercise 7.2: High Value Orders
-- **Task**: Find all orders with a total amount greater than 200.
```

```
-- **Expected Result**: 8 orders


select 'Exercise 7.2:' as "";
select * from orders where total_amount>200;
-- ---
```

```
TERMINAL

Exercise 7.2:
order_id        customer_name   product_id      quantity        order_date      total_amount    status
1       Amit    1       3       2024-03-05      300.00  Processing
2       Bina    2       2       2024-03-10      250.00  Shipped
3       Chetan  3       2       2024-03-15      220.00  Delivered
4       Divya   4       1       2024-03-20      210.00  Processing
5       Esha    5       2       2024-02-12      260.00  Delivered
6       Farhan  6       1       2024-02-25      230.00  Pending
7       Gopal   7       2       2024-04-02      240.00  Shipped
8       Harish  8       1       2024-01-18      260.00  Pending
```

```
-- ### Exercise 7.3: Order Status
-- **Task**: Find all orders that are currently being processed or shipped.
-- **Expected Result**: 4 orders


select 'Exercise 7.3:' as "";
select * from orders where status='Processing' or status='Shipped';
```

```
TERMINAL

Exercise 7.3:
order_id        customer_name   product_id      quantity        order_date      total_amount    status
1       Amit    1       3       2024-03-05      300.00  Processing
2       Bina    2       2       2024-03-10      250.00  Shipped
4       Divya   4       1       2024-03-20      210.00  Processing
7       Gopal   7       2       2024-04-02      240.00  Shipped
```

```
-- ### Exercise 7.4: Cancelled Orders
-- **Task**: Find all cancelled orders.
-- **Expected Result**: 1 order


select 'Exercise 7.4:' as "";
select * from orders where status='Cancelled';
```

```
TERMINAL

Exercise 7.4:
order_id        customer_name   product_id      quantity        order_date      total_amount    status
10      Jay     10      1       2024-04-05      100.00  Cancelled
```

```
-- ## Part 8: Complex Queries
-- ### Exercise 8.1: Department Analysis
-- **Task**: Find all IT employees in San Francisco who earn more than 80000.
-- **Expected Result**: 3 employees


select 'Exercise 8.1:' as "";
select * from employees where department='IT' and city='San Francisco' and
salary>80000;
```

```
TERMINAL                                                                wsl -

Exercise 8.1:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
```

```
-- ### Exercise 8.2: Hiring Date Range
-- **Task**: Find all employees hired between January 1, 2020 and December 31,
2021.
-- **Expected Result**: 9 employees


select 'Exercise 8.2:' as "";
select * from employees where hire_date between '2020-01-01' and '2021-12-31';
```

```
TERMINAL

Exercise 8.2:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
12      Sneha   Anderson        sneha@company.com       HR      Exec    52000   2021-01-05      11 Delhi
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ### Exercise 8.3: Email Domain Search
-- **Task**: Find all employees whose email contains 'company.com'.
-- **Expected Result**: 15 employees (all employees)


select 'Exercise 8.3:' as "";
select * from employees where email like '%company.com%';
```

```
TERMINAL

Exercise 8.3:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date   manager_id      city
1       John    Sharma  john@company.com        IT      Dev     90000   2020-01-10      NULL    San Francisco
2       James   Johnson james@company.com       IT      Dev     95000   2019-02-12      1       San Francisco
3       Jack    Singh   jack@company.com        IT      Lead    92000   2019-03-15      1       San Francisco
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
5       Nea     Gupta   neha@company.com        IT      Tester  72000   2019-05-20      1       Chicago
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
7       Pooja   Mehta   pooja@company.com       Sales   Mgr     85000   2019-07-12      6       Delhi
8       Suresh  Rao     suresh@company.com      Sales   Exec    55000   2020-08-12      6       New York
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
12      Sneha   Anderson        sneha@company.com       HR      Exec    52000   2021-01-05      11 Delhi
13      Vikas   Jain    vikas@company.com       Finance Acc     61000   2021-02-10      1       Chicago
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
15      Anita   Roy     anita@company.com       Finance Rep     48000   2021-04-15      1       New York
```

```
-- ### Exercise 8.4: Product Inventory Alert
-- **Task**: Find all Electronics products that either have zero stock or less
than 30 units, and price above 50.
-- **Hint**: Use parentheses to group the OR condition for stock
-- **Expected Result**: 2 products


select 'Exercise 8.4:' as "";
select * from products where category='Electronics' and (stock_quantity=0 or
stock_quantity<30) and price>50;
```

```
TERMINAL

Exercise 8.4:
product_id      product_name    category        price   stock_quantity  supplier
9       Pen Drive       Electronics     600.00  20      TechCorp
14      Calculator      Electronics     900.00  10      TechCorp
```

```
-- ---
-- ### Exercise 8.5: Multi-Condition Order Search
-- **Task**: Find all delivered orders from February or March 2024 with total
amount over 150.
-- **Expected Result**: 2 orders


select 'Exercise 8.5:' as "";
select * from orders where status='Delivered' and order_date between
'2024-02-01' and '2024-03-31' and total_amount>150;
```

```
g Console (Ctrl+Shift+Y)

Exercise 8.5:
order_id        customer_name   product_id      quantity        order_date      total_amount    status
3       Chetan  3       2       2024-03-15      220.00  Delivered
5       Esha    5       2       2024-02-12      260.00  Delivered

Challenge 9.1:
```

```
-- ## Part 9: Challenge Exercises
-- ### Challenge 9.1: Salary Band Analysis
-- **Task**: Categorize employees into salary bands. Find all employees earning
between 50000-60000 and label them as 'Entry Level'.
```

```
-- **Hint**: Use a string literal in SELECT to add the label
-- **Expected Result**: Employees in entry level salary band


select 'Challenge 9.1:' as "";
select first_name,last_name,salary,'Entry Level' as level from employees where
salary between 50000 and 60000;
```

```
TERMINAL

Challenge 9.1:
first_name      last_name       salary  level
Rohit   Wilson  60000   Entry Level
Suresh  Rao     55000   Entry Level
Ritika  Nair    60000   Entry Level
Manoj   Iyer    58000   Entry Level
Sneha   Anderson        52000   Entry Level
Arun    Malik   60000   Entry Level
on
```

```
-- ### Challenge 9.2: Name Pattern Search
-- **Task**: Find employees whose first name contains 'ar' OR last name starts
with 'W' OR 'T'.
-- **Expected Result**: 3 employees


select 'Challenge 9.2:' as "";
select * from employees where first_name like '%ar%' or last_name like 'W%' or
last_name like 'T%';
```

```
TERMINAL

Challenge 9.2:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date       manager_id      city
4       Julia   Thomas  julia@company.com       IT      Tester  72000   2019-04-18      1       New York
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
14      Arun    Malik   tarun@company.com       Finance Acc     60000   2021-03-12      13      Boston
```

```
-- ---
-- ### Challenge 9.3: Product Reorder List
-- **Task**: Create a reorder list for products that are:
-- - In stock (quantity > 0)
-- - Low stock (quantity < 50)
-- - Price under 100
-- - From specific suppliers (TechCorp or CableWorks)
-- **Expected Result**: Products matching all criteria



select 'Challenge 9.3:' as "";
select * from products where stock_quantity>0 and stock_quantity<50 and
price<100 and (supplier='TechCorp' or supplier='CableWorks');

-- ---
```

```
Challenge 9.3:
product_id      product_name    category        price   stock_quantity  supplier
4       Mouse   Electronics     80.00   35      CableWorks
7       Charger Electronics     90.00   30      TechCorp
  Boston
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULLBoston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9  Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
```

```
-- ### Challenge 9.4: Employee Location and Department
-- **Task**: Find employees who:
-- - Work in Sales, Marketing, or HR
-- - Are located in cities that start with 'C' or 'B'
-- - Have a salary not equal to 54000 or 52000
-- **Expected Result**: 4 employees



select 'Challenge 9.4:' as "";
select * from employees where (department='Sales' or department='Marketing' or
department='HR') and (city like 'C%' or city like 'B%') and salary<>54000 and
salary<>52000;
```

```
Challenge 9.4:
employee_id     first_name      last_name       email   department      job_title       salary  hire_date       manager_id      city
6       Rohit   Wilson  rohit@company.com       Sales   Exec    60000   2019-06-10      NULL    Boston
9       Kiran   Das     kiran@company.com       Marketing       Mgr     74000   2020-09-15      NULL    Boston
10      Ritika  Nair    ritu@company.com        Marketing       Exec    60000   2020-10-18      9       Chicago
11      Manoj   Iyer    manoj@company.com       HR      Mgr     58000   2020-11-20      1       Boston
```

```
-- ### Challenge 9.5: Order Investigation
-- **Task**: Find orders where:
-- - Customer name starts with a letter between 'A' and 'H'
-- - Order date is NOT in January 2024
-- - Status is NOT cancelled
-- - Quantity ordered is between 2 and 6
-- **Hint**: Use string comparison for customer name filter
-- **Expected Result**: 5 orders


select 'Challenge 9.5:' as "";
select * from orders where customer_name>='A' and customer_name<='H' and
order_date not between '2024-01-01' and '2024-01-31' and status<>'Cancelled'
and quantity between 2 and 6;
```

```
Challenge 9.5:
order_id        customer_name   product_id      quantity        order_date      total_amount    status
1       Amit    1       3       2024-03-05      300.00  Processing
2       Bina    2       2       2024-03-10      250.00  Shipped
3       Chetan  3       2       2024-03-15      220.00  Delivered
5       Esha    5       2       2024-02-12      260.00  Delivered
7       Gopal   7       2       2024-04-02      240.00  Shipped
```

```
-- ---
-- ## Self-Assessment Checklist
-- After completing the lab, check off each skill you feel confident with:
-- - [ ] Writing basic SELECT statements
-- - [ ] Selecting specific columns
-- - [ ] Using column aliases
-- - [ ] Using SELECT DISTINCT
-- - [ ] Performing arithmetic operations in SELECT
-- - [ ] Concatenating strings
-- - [ ] Using WHERE clause with comparison operators
-- - [ ] Using AND, OR, NOT logical operators
-- - [ ] Using BETWEEN operator
-- - [ ] Using IN operator
-- - [ ] Using LIKE operator with wildcards
-- - [ ] Using IS NULL and IS NOT NULL
-- - [ ] Combining multiple conditions with proper parentheses
-- - [ ] Writing complex queries with multiple tables
-- - [ ] Troubleshooting SQL syntax errors
-- ## Tips for Success
-- 1. **Test incrementally**: Start with simple queries and add complexity
gradually
-- 2. **Check row counts**: Verify your results match the expected number of
rows
-- 3. **Use proper formatting**: Indent your queries for better readability
-- 4. **Remember quotes**: Text values need single quotes, numbers don't
-- 5. **NULL is special**: Use IS NULL, not = NULL
-- 6. **Parentheses matter**: Use them to control logic in complex conditions
-- 7. **LIKE is case-sensitive**: Depending on your database configuration
-- 8. **Ask questions**: If you're stuck, ask your instructor for guidance
-- ## Common Mistakes to Avoid
-- 1. Forgetting quotes around text values
-- 2. Using = instead of IS NULL
-- 3. Missing parentheses in complex conditions
-- 4. Incorrect wildcard usage (% vs _)
-- 5. Wrong date format
-- 6. Confusing AND/OR precedence
-- ## Notes Section
-- Use this space to write down important points, questions, or observations:
-- ---
-- ---
-- ---
-- ---
-- ---
```